EE 3072
System Dynamics and Control

Lecture 1

Introduction to Control Systems

Outline

- What is a control system?
- A brief history of control
- Basic components of a control system
- Open-loop control vs. closed-loop control
- Classification of control systems
- Basic requirements of control systems
- Summary

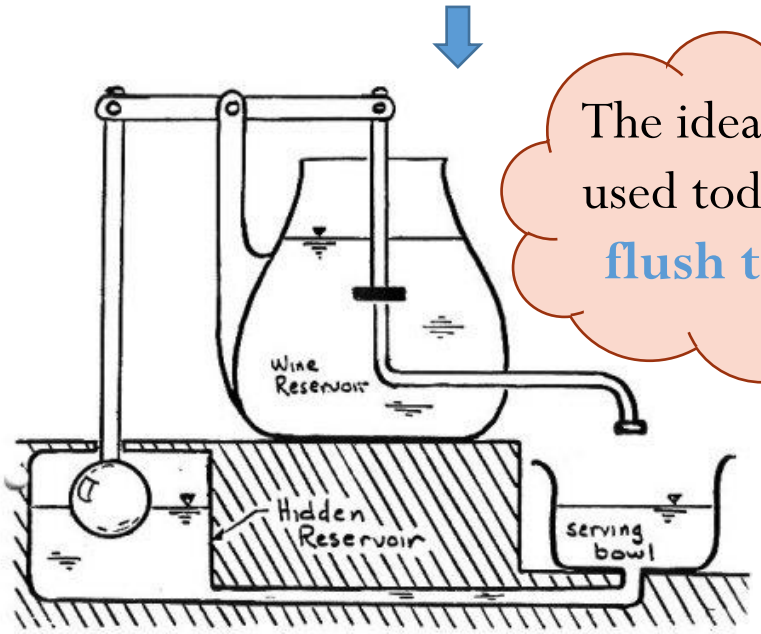
What is a control system?

- Generally speaking, a control system is a system that is used to realize a desired output or objective.

- Control systems are everywhere
  - They appear in our homes, in cars, in industry, in scientific labs, in hospitals …
  - Principles of control have an impact on diverse fields as engineering, aeronautics, economics, biology and medicine …
  - Wide applicability of control has many advantages (e.g., it is a good vehicle for technology transfer)

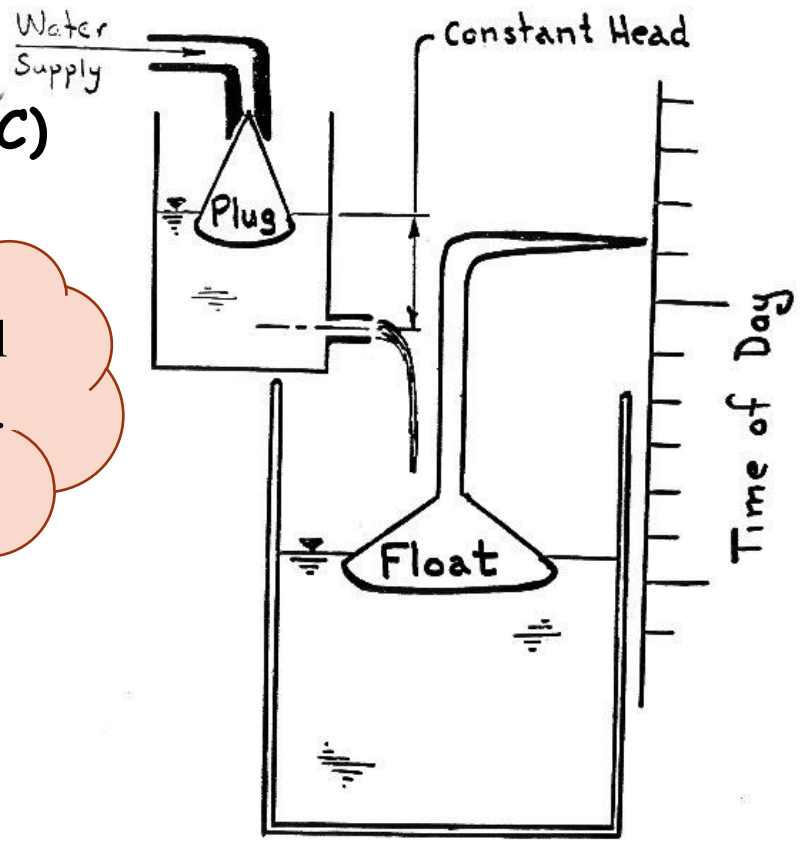
A brief history of control

- **Two of the earliest examples**
  - **Water clock (270 BC)**
  - **Self-leveling wine vessel (100BC)**



HERO'S SELF-LEVELING BOWL
ca. 30 B.C.

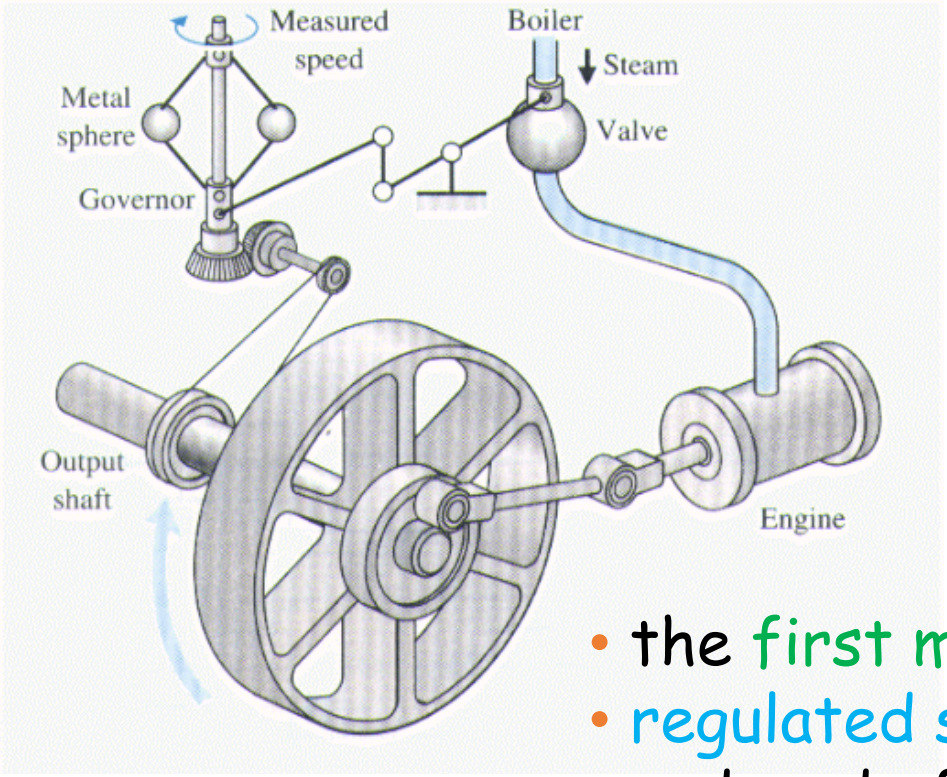
The idea is still used today, i.e. **flush toilet**



an ELEMENTARY WATER CLOCK

A brief history of control

- Fly-ball governor (James Watt, 1769)





- the first modern controller
- regulated speed of steam engine
- reduced effects of variances in load
- propelled Industrial Revolution

A brief history of control

- Birth of mathematical control theory
  - G. B. Airy (1840)
    - the first one to discuss instability in a feedback control system
    - the first to analyze such a system using differential equations
  - J. C. Maxwell (1868)
    - the first systematic study of the stability of feedback control
  - E. J. Routh (1877)
    - deriving stability criterion for linear systems
  - A. M. Lyapunov (1892)
    - deriving stability criterion that can be applied to both linear and nonlinear differential equations
    - results not introduced in control literature until about 1958

A brief history of control

- Birth of classical control design method
  - H. Nyquist (1932)
    - developed a relatively simple procedure to determine stability from a graphical plot of the loop-frequency response.
  - H. W. Bode (1945)
    - frequency-response method
  - W. R. Evans (1948)
    - root-locus method

core of classical control design

With the above methods, we can design control systems that are stable, acceptable but not necessarily optimal in some pre-specified sense.

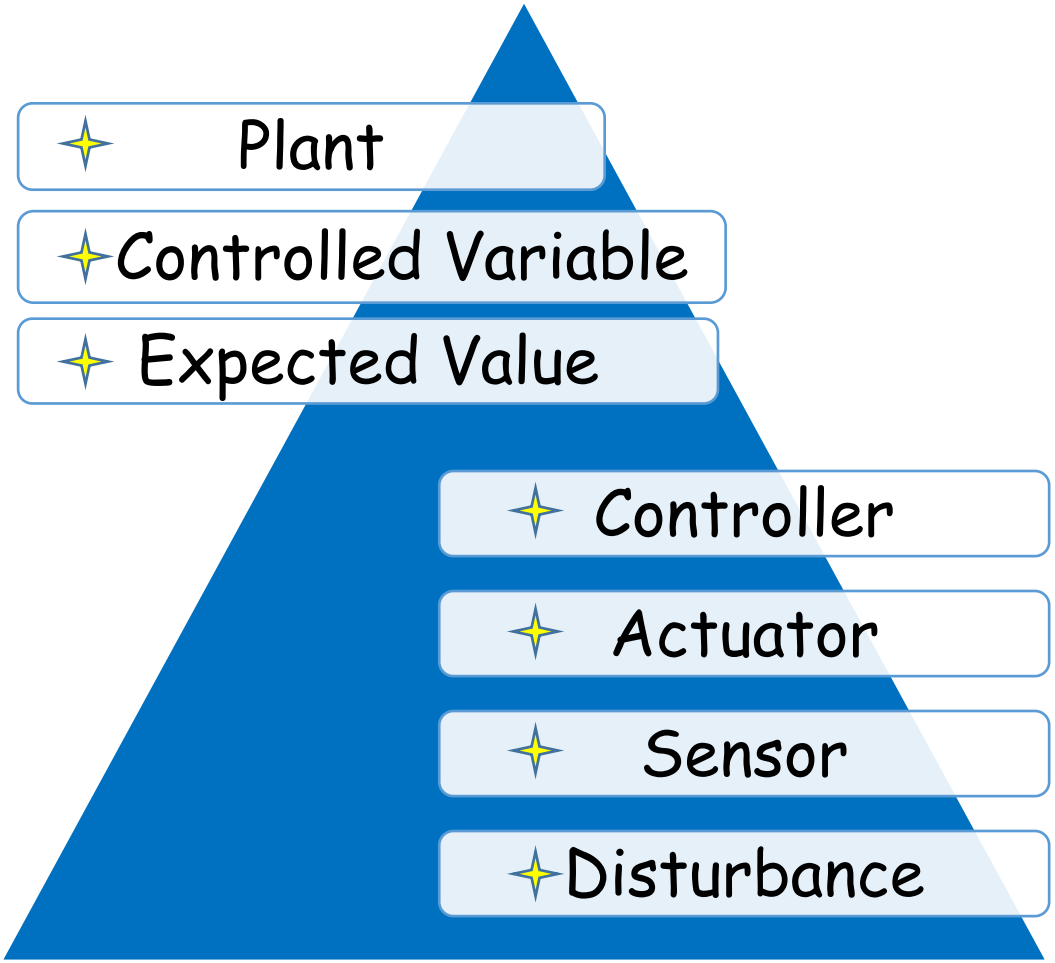
A brief history of control

- Development of modern control design
  - Late 1950s: designing optimal systems in some meaningful sense
  - 1960s: digital computers help time-domain analysis of complex systems, modern control theory has been developed to cope with the increased complexity of modern plants
  - 1960s~1980s: optimal control of both deterministic and stochastic systems; adaptive control and learning control
  - 1980s~present: robust control, H-inf control…

A brief history of control

- Recent applications of modern control theory include such non-engineering systems as biological, biomedical, economic and socioeconomic systems …

Basic components of a control system



Basic concepts of a control system



1.Plant: a physical object to be controlled such as a mechanical device, a heating furnace, a chemical reactor or a spacecraft.



2.Controlled variable: the variable controlled by Automatic Control System, generally refers to the system output.



3.Expected value: the desired value of controlled variable based on requirement, often it is used as the reference input.

Controller

4. Controller: an agent that can calculate the required control signal.

Actuator

5. Actuator: a mechanical device that takes energy, usually created by air, electricity, or liquid, and converts that into form suitable for the plant.

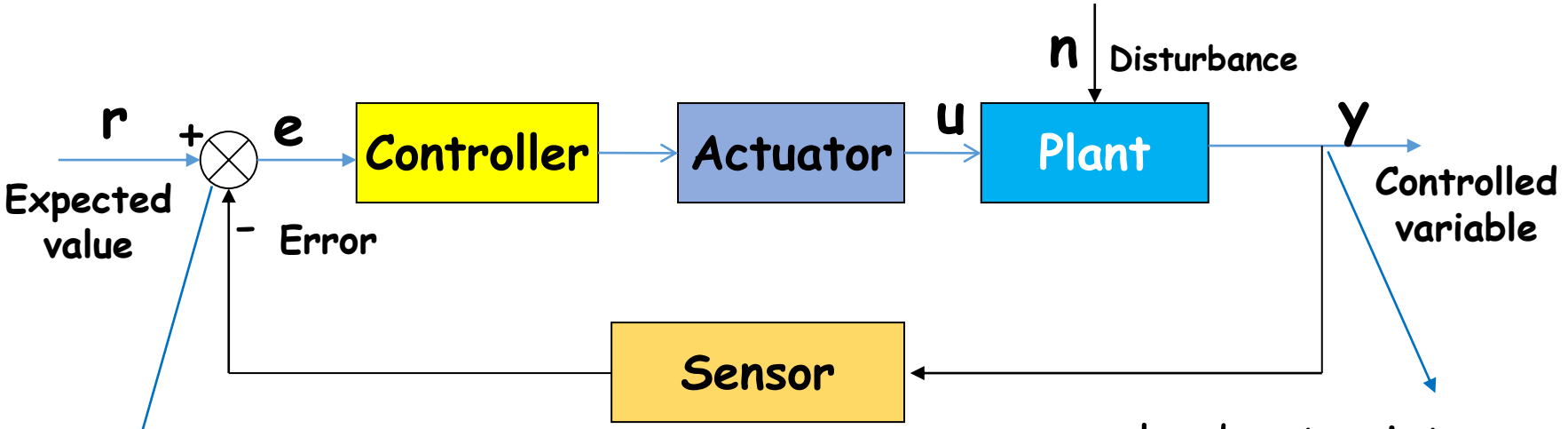
Sensor

6. Sensor: a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an instrument.

Disturbance

7. Disturbance: the unexpected factors disturbing the normal functional relationship between the controlling and controlled parameter variations.

Block diagram of a control system



comparison component (comparison point) : its output equals the algebraic sum of all input signals.

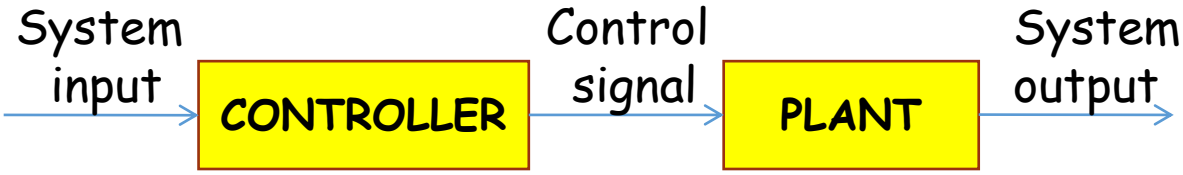
"+": plus; "-": minus

The Block represents the function and name of its corresponding mode, we don't need to draw detailed structure, and the line guides for the transfer route.

lead-out point: Here, the signal is transferred along two separate routes.

Open-loop control systems

- Open-loop control systems: those systems in which the output has no effect on the control action.



- The output is neither measured nor fed back for comparison with the input.
- For each reference input, there corresponds a fixed operating conditions; the accuracy of the system depends on *calibration*.
- In the presence of *disturbances*, an open-loop system will not perform the desired task.

Open-loop control systems

- Examples
  - Washing machine



  - Traffic signals



Note that any control systems that operates on a time basis are open-loop.

Open-loop control systems

- Some comments on open-loop control systems
  - Simple construction and ease of maintenance.
  - Less expensive than a closed-loop system.

Good

  - Recalibration is necessary from time to time.
  - Sensitive to disturbances and uncertainties, so less accurate.

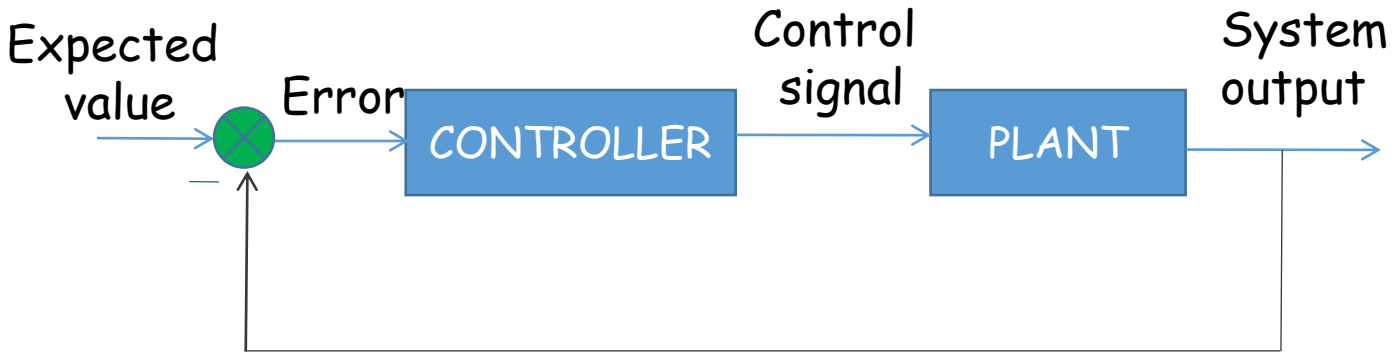
Bad

Open-loop control systems

- When should we apply open-loop control?
  - The relationship between the input and output is *exactly known*.
  - There are *neither* internal *nor* external disturbances.
  - Measuring the output precisely is *very hard* or *economically infeasible*.

Closed-loop control systems

- Closed-loop control systems are often referred to as feedback control systems.
- The idea of feedback:
  - Compare the *actual output* with the *expected value*.
  - Take actions based on the *difference (error)*.



  - This seemingly simple idea is tremendously powerful.
  - Feedback is *a key idea* in the discipline of control.

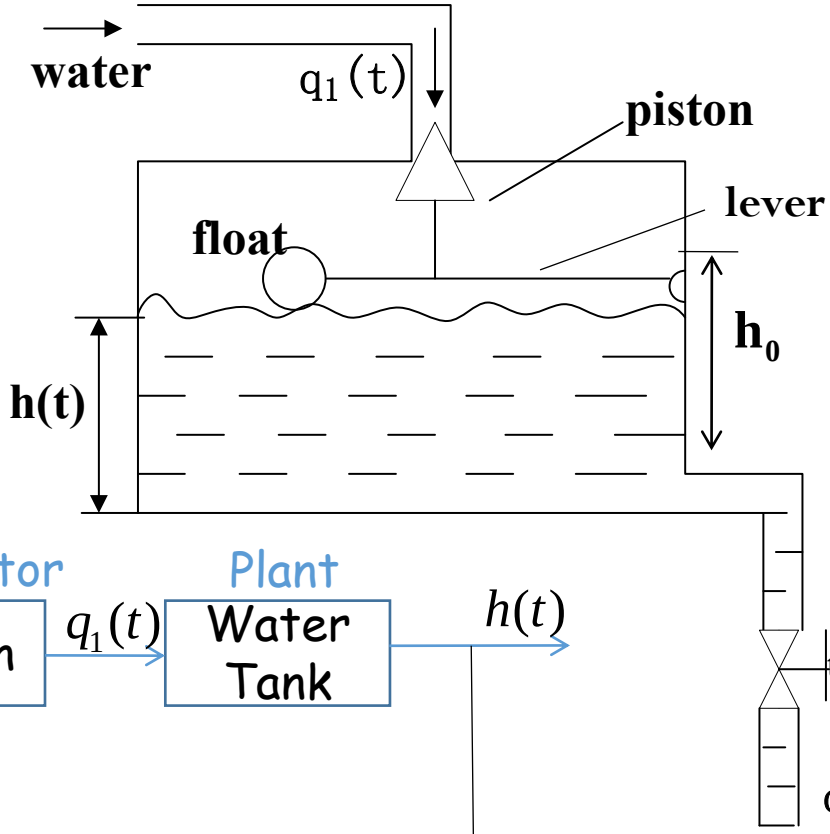
Closed-loop control systems

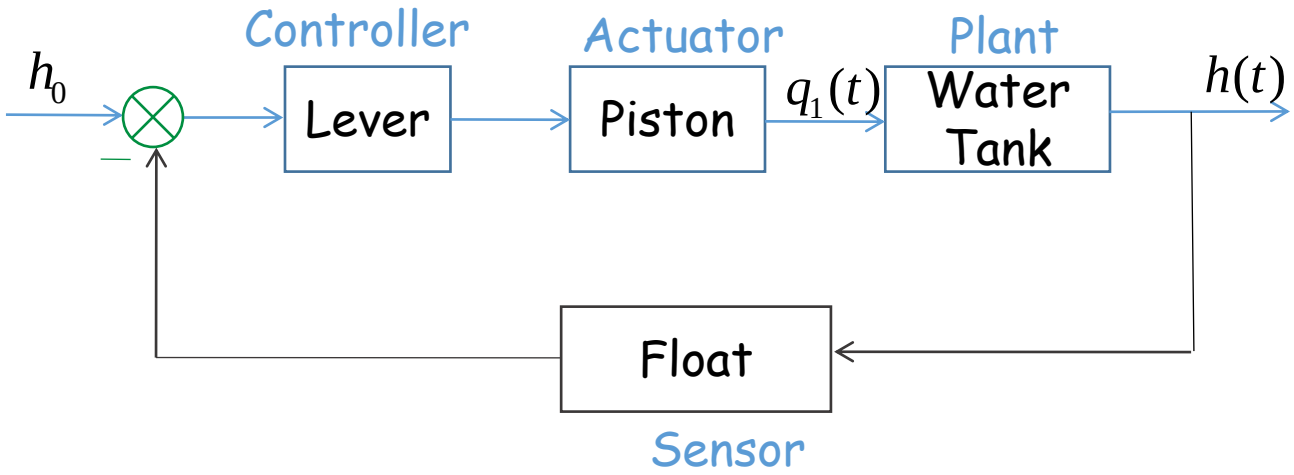
- In practice, feedback control system and closed-loop control system are used interchangeably

- Closed-loop control always implies the use of feedback control action in order to reduce system error

Example 1: flush toilet

Plant: water tank
Input: water flow
Output: water level $h(t)$
Expected value: h_0
Sensor: float
Controller: lever
Actuator: piston



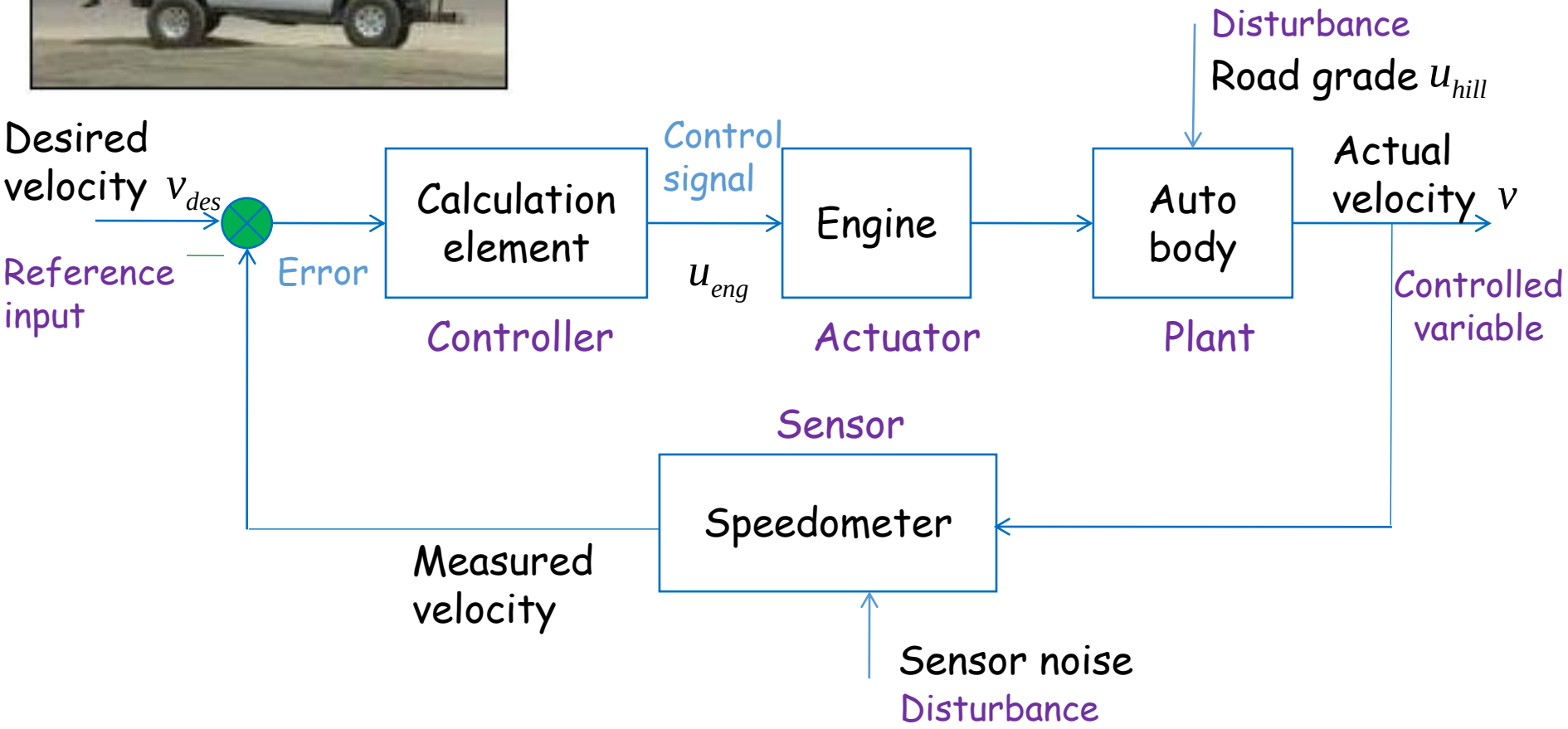


Example 2: Cruise control



$$m\dot{v} = -bv + u_{eng} + u_{hill}$$

$$u_{eng} = k(v_{des} - v)$$

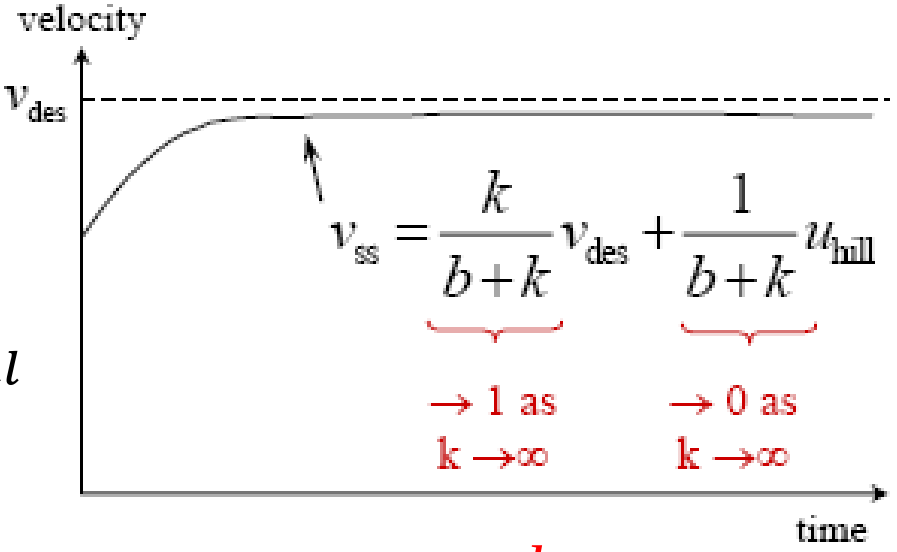


Example 2: Cruise control



$$m\dot{v} = -bv + u_{eng} + u_{hill}$$

$$u_{eng} = k(v_{des} - v)$$



$$v_{ss} \to v_{des} \ as \ k \to \infty$$

Stability/performance

Steady state velocity approaches desired velocity as $k \to \infty$;
Smooth response: no overshoot or oscillations

Disturbance rejection

Effect of disturbances (eg, hills) approaches zero as $k \to \infty$

Robustness

Results don't depend on the specific values of *b*, *m* or *k*, *for k sufficiently large*

Note

- In this example, we ignore the dynamic response of the car and consider only the steady behavior.
  - Dynamics will play a major role in later chapters.
- There are limits on how high the gain k can be made.
  - when dynamics are introduced, the feedback can make the response worse than before, or even cause the system to be unstable.

Comments on feedback control

- Main advantages of feedback:
  - reduce disturbance effects
  - make system insensitive to variations
  - stabilize an unstable system
  - create well-defined relationship between output and reference

Comments on feedback control

- Potential drawbacks of feedback:
  - may cause instability if not used properly
  - couple noise from sensors into the dynamics of a system
  - increase the overall complexity of a system

Comments on feedback control

- Feedback control design:

  how to

  get the gain as large as possible to reduce the error

  without

  making the system become unstable.

Other examples of feedback

The human body is highly advanced feedback control system.

Body temperature and blood pressure are kept constant by means of physiological feedback.

Feedback makes the human body relatively insensitive to external disturbance. Thus we can survive in a changing environment.

Open-loop vs. closed-loop

- Open-loop control

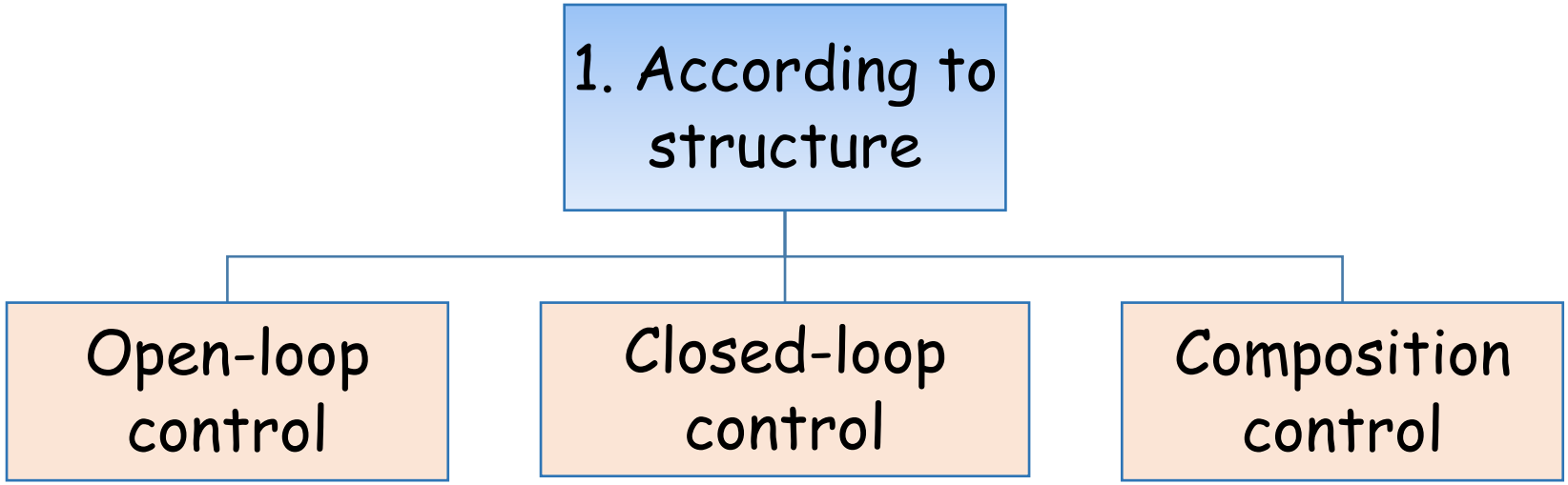
  - Simple structure, low cost
  - Easy to regulate
  - Low accuracy and resistance to disturbance

- Closed-loop control

  - Ability to correct error
  - High accuracy and resistance to disturbance
  - Complex structure, high cost
  - Selecting parameter is critical (may cause stability problem)

Open-loop + Closed-loop = Composite control system

Classification of control systems



Classification of control systems

2. According to reference input

Constant-value control

- the reference input (expected value) is a constant value
- the controller works to keep the output around the constant value

  e.g. constant-temperature control, liquid level control and constant-pressure control.

Servo/tracking control

- the reference input may be unknown or varying
- the controller works to make the output track the varying reference

  e.g. automatic navigation systems on boats and planes, satellite-tracking antennas

Programming control

- the input changes according to a program
- the controller works according to predefined command

  e.g. numerical control machine

Classification of control systems

3. According to system characteristics

- Linear control system
- Nonlinear control system

$$f(x_1) = y_1 \qquad f(x_2) = y_2$$

superposition principle

$$f(\alpha_1 x_1 + \alpha_2 x_2) = f(\alpha_1 x_1) + f(\alpha_2 x_2) = \alpha_1 y_1 + \alpha_2 y_2$$

- superposition principle applies
- described by linear differential equation

- described by nonlinear differential equation

Remark on nonlinear systems

- Quite often, nonlinear characteristics are intentionally introduced in a control system to improve its performance or provide more effective control.

  For instance, to achieve minimum-time control, an on-off (bang-bang or relay) type controller is used in many missile or spacecraft control systems

- There are no general methods for solving a wide class of nonlinear systems

Classification of control systems

4. According to signal form

- Continuous time control system

  All the signals are functions of continuous time variable t

- Discrete time control system

  Signals are in the form of either a pulse train or a digital code

  e.g. digital control system

Remark on digital control systems

- A digital control system refers to the use of a digital computer or controller in the system, so that the signals are digitally coded, such as in binary code.
- Digital computers provide many advantages in size and flexibility.
  - The expensive equipment used in a system may be shared simultaneously among several control channels.
  - Digital control systems are usually less sensitive to noise.

Classification of control systems

5. According to parameters

- Time-invariant system

  The parameters of a control system are stationary with respect to time

- Time-varying system

  System contains elements that drift or vary with time

  e.g. Guided-missile control system, time-varying mass results in time-varying parameters of the control system

Basic requirements for control systems

- **Stability**: refers to the ability of a system to recover equilibrium
- **Quickness**: refers to the duration of transient process before the control system reaches its equilibrium
- **Accuracy**: refers to the size of steady-state error when the transient process ends

  (Steady-state error=desired output – actual output)

Note

- For a control system, the above three performance indices (stability, quickness, accuracy) are sometimes contradictory.

- In design of a practical control system, we always need to make compromise.

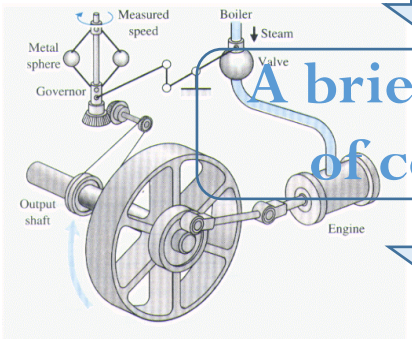
Review questions

1. A closed-loop control system is usually more accurate than an open-loop system. (T) (F)
2. Feedback is sometimes used to improve the sensitivity of a control system. (T) (F)
3. If an open-loop system is unstable, then applying feedback will always improve its stability. (T) (F)
4. Feedback can cause instability. (T) (F)
5. Nonlinear elements are sometimes intentionally introduced to a control system to improve its performance. (T) (F)
6. Discrete-data control systems are more susceptible to noise due to the nature of its signals. (T) (F)

Summary

What is a control system?

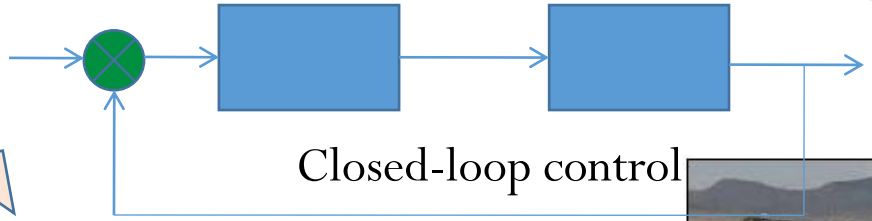
A brief history of control





Open-loop control



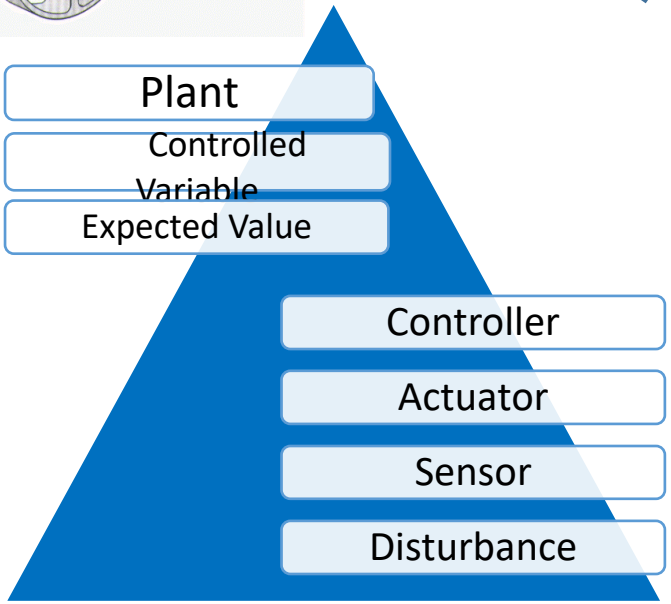


Closed-loop control

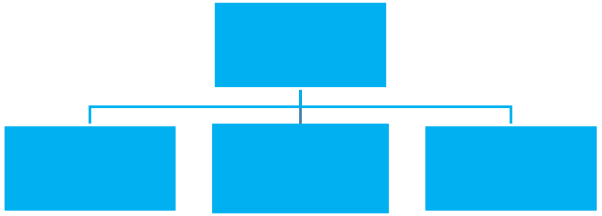


Feedback:

- reduce disturbance effects
- provide extra freedom for control design
- may cause instability



Classification of control systems



Requirements: Stability, Quickness, Accuracy

Some suggestions on course study

- Pay enough attention during lecture
  - Control theory is very interesting, very useful, but sometimes, very difficult.
- Study regularly the covered topics
- Regularly solve example problems - practice, practice and practice